Inflectional class phonotactics affect lexical access

Inflectional classes (IC) are often idiosyncratic. For example, in Spanish, verbal stems belong to one of three classes that determine the form of their infinitival suffix: *-ar*, *-er*, or *-ir*. Given near-minimal pairs of stems across the classes, e.g. *sed-ar* ('to sedate'), *ced-er* ('to cede'), and *ped-ir* ('to ask'), it is clear that the phonological form of the stem cannot categorically predict the form of the suffix. Despite this surface arbitrariness, recent research has shown that speakers are able to form statistically significant phonotactic generalizations over morpho-phonologically defined subsets of the lexicon (e.g. [1-3]), including ICs [4]. However, this work uses metalinguistically loaded forced-choice nonce word judgement tasks—making little predictions about whether or how these generalizations should affect real time processing. *This study tests whether such IC generalizations affect the lower level processing stage of lexical access.*

This study: I will first establish that there are significant differences in stem final vowel heights across the three Spanish verbal inflectional classes (tab. 1), as suggested by historic shifts [5]. I then test whether conforming or violating this generalization affects responses in a visual lexical decision task (LDT). Since responses are directly related to (un)successful access of lexical items, any behavioral effects of inflectional class phonotactics can be attributed to lexical access. If these phonotactic generalizations are tied to the mental lexicon, then: 1) real stems that conform to the generalization should be easier to accept; 2) nonce stems that conform to a generalization should be harder to reject. Otherwise, if these generalizations are a product of down-stream nonce word evaluation, no effects are expected for real stems.

Simulation Methods: Stem final vowel height was quantified over verbal lemmas taken from the cleaned Corpus del Español [6]. To test whether ICs differed in their vowel height, a Monte-Carlo simulation was computed for each class and vowel height [7].

Simulation Results: Results are given in fig. 1. The IR class showed a significant skew towards -high vowels (e.g. *v**i**v-ir*, 'to live') and away from +high vowels (e.g. ***a**br-ir*, 'to open'). The ER class showed the opposite effect, with a skew towards -high vowels (e.g. *c**o**m-er*, 'to eat'). While the corpus had 1,068 +high ER stems, upon manual inspection all were noise (e.g. *copywriter*). While the AR class showed a significant skew towards +high vowels (e.g. ***pi**c-ar,* 'to chop'), it was a proportionally smaller effect and may simply be a product of noise in the corpus.

LDT Methods: Response time (RT) and accuracy (Acc) were taken from responses to a visual LDT megastudy [8]. Real and nonce verbs were analyzed separately. Linear models, for RT, and Logistic models, for Acc, were fit with main effects for average bigram freq., stimulus length, log lemma freq., and an interaction term of IC and vowel height (tab. 2-3). Of interest were the Estimated marginal means (EMM) for each interaction term factor (tab. 4-5).

LDT Results: Main effects for class were significant for both nonce and real word datasets for both RT and Acc—participants were treating these items as verbs. For nonce words, EMMs revealed significant effects of vowel height in opposite directions for the IR and ER classes, confirming the hypothesis that *nonce words that look like they belong to their class are harder and slower to reject*. For real words, EMMs revealed a significant effect of vowel height on accuracy for the IR class, confirming the hypothesis that *real verbs that look like they do not belong to their class are harder to accept*. The AR class showed the same directionality of effect as the main effect for vowel height regardless of lexicality—likely reflecting a lexicon wide effect of vowel height. Despite the simplicity of the models, these results clearly show that IC-level phonotactic generalizations affect responses for both real and nonce words—suggesting that they play a role in both lexical access and the computations underpinning the mental lexicon.

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[1] Becker, & Gouskova (2016) [2] Becker, Ketrez, & Nevins (2011) [3] Gouskova, Newlin-Łukowicz, & Kasyanenko (2015) [4] Tabachnick (2023) [5] Montgomery (1976) [6] Davies (2013) [7] Gouskova (2025) [8] Aguasvivas et al. (2018)

Simulation Results

	+ High	-High
AR	pic-ar *'to chop'*	am-ar *'to love'*
ER	-	tañ-er *'to strum'*
IR	viv-ir *'to live'*	abr-ir *'to open'*

Table 1: Examples of Vowel Heights across Spanish verbs

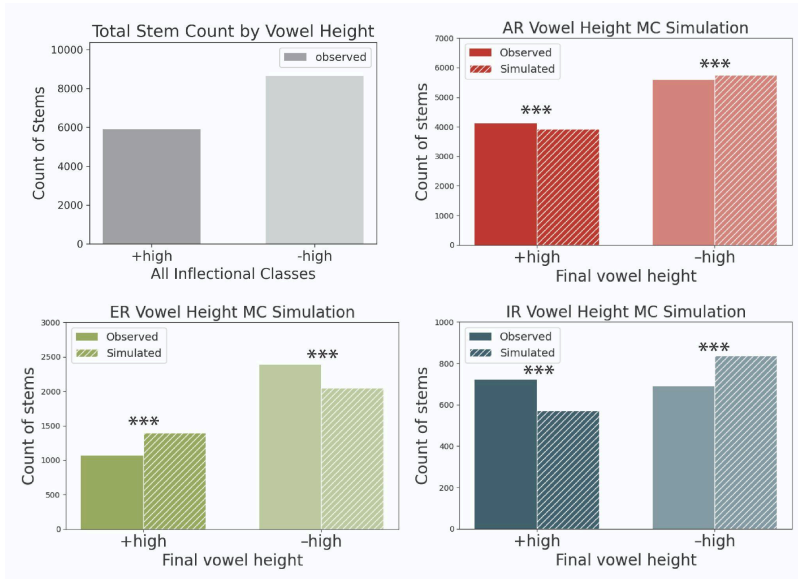

fig. 1: Monte-Carlo Simulations for three ICs. Average of simulation given in striped. Observed values given in solid.

Visual Lexical Decision Task Results

Nonce Words

	Accuracy		Response Time	
Variable	Estimate	P-Val	Estimate	P-Val
intercept	2.942680	<0.0001 ***	1244	<0.0001 ***
ER-class	0.883365	<0.0001 ***	-50.8	<0.0001 ***
IR-class	0.505625	<0.0001 ***	-29.7	<0.0001 ***
High V	0.042494	<0.0001 ***	-15.4	<0.0001 ***
Avg. Bigram Freq.	-0.031834	<0.0001 ***	1.614	<0.0001 ***
Length	0.300697	<0.0001 ***	68.0	<0.0001 ***
ER-class: High-V	0.224370	<0.0001 ***	-32.8	<0.0001 ***
IR-class: High-V	-0.186894	<0.0001 ***	30.9	<0.0001 ***

table 2: Main effects for accuracy and RT models for nonce words. Accurate response is correctly rejecting the nonce word.

Real Words

	Accuracy		Response Time	
Variable	Estimate	P-Val	Estimate	P-Val
intercept	2.942680	<0.0001 ***	1244	<0.0001 ***
ER-class	0.883365	<0.0001 ***	-50.8	<0.0001 ***
IR-class	0.505625	<0.0001 ***	-29.7	<0.0001 ***
High V	0.042494	<0.0001 ***	-15.4	<0.0001 ***
Avg. Bigram Freq.	-0.031834	<0.0001 ***	1.614	<0.0001 ***
Length	0.300697	<0.0001 ***	68.0	<0.0001 ***
ER-class: High-V	0.224370	<0.0001 ***	-32.8	<0.0001 ***
IR-class: High-V	-0.186894	<0.0001 ***	30.9	<0.0001 ***

table 3: Main effects for accuracy and RT models for real words. Accurate response is correctly accepting the real word.

IC	Accuracy		RT	
	Estimate	P-val	Estimate	P-val
AR	-0.05	<0.0001	23.63	<0.0001
ER	NA	NA	NA	NA
IR	-0.20	<0.0001	1.40	1.0

table 4: Expanded interaction terms for nonce words from EMMs *

IC	Accuracy		RT	
	Estimate	P-val	Estimate	P-val
AR	-0.04	0.0001	15.38	0.0001
ER	-0.27	<0.0001	48.20	<0.0001
IR	0.14	<0.0001	-15.55	<0.0001

table 5: Expanded interaction terms for real words from EMMs *

**Accuracy: Positive value indicates higher accuracy for -high, negative value indicates lower accuracy for -high.*

RT: positive value indicates slower RT for -high, negative value indicates faster RT for -high.